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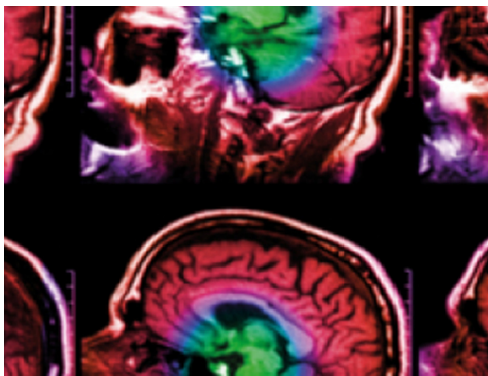
Spatially fractionated radiotherapy: tumor response modelling including immunomodulation

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PAPER

Spatially fractionated radiotherapy: tumor response modelling including immunomodulation

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24 August 2021**Abstract**

A mathematical tumor response model has been developed, encompassing the interplay between immune cells and cancer cells initiated by either partial or full tumor irradiation. The iterative four-compartment model employs the linear–quadratic radiation response theory for four cell types: active and inactive cytotoxic T lymphocytes (immune cells, CD8⁺ T cells in particular), viable cancer cells (undamaged and reparable cells) and doomed cells (irreparably damaged cells). The cell compartment interactions are calculated per day, with total tumor volume (TV) as the main quantity of interest. The model was fitted to previously published data on syngeneic xenografts (67NR breast carcinoma and Lewis lung carcinoma; (Markovsky *et al* 2019 *Int. J. Radiat. Oncol. Biol. Phys.* **103** 697–708)) subjected to single doses of 10 or 15 Gy by 50% (partial) or 100% (full) TV irradiation. The experimental data included effects from anti-CD8⁺ antibodies and immunosuppressive drugs. Using a new optimization method, promising fits were obtained where the lowest and highest root-mean-squared error values were observed for anti-CD8⁺ treatment and unirradiated control data, respectively, for both cell types. Additionally, predictive capabilities of the model were tested by using the estimated model parameters to predict scenarios for higher doses and different TV irradiation fractions. Here, mean relative deviations in the range of 19%–34% from experimental data were found. However, more validation data is needed to conclude on the model's predictive capabilities. In conclusion, the model was found useful in evaluating the impact from partial and full TV irradiation on the immune response and subsequent tumor growth. The model shows potential to support and guide spatially fractionated radiotherapy in future pre-clinical and clinical studies.

1. Introduction

Spatially fractionated radiotherapy (SFRT) has emerged as a promising alternative to conventional radiotherapy (RT) (Mohiuddin *et al* 1999, Billena and Khan 2019, Yan *et al* 2020). Departing from the paradigm where the entire tumor should be irradiated with a homogeneous dose, SFRT involves the use of a highly non-uniform dose distribution. Two-dimensional (2D) orthovoltage X-ray therapy using multi-hole collimators was the first form of SFRT and was developed as a means to reduce radiation toxicity by delivering locally high doses in a hypofractionated setting (Laissue *et al* 2012). Current SFRT using megavoltage X-rays can be delivered both in 2D ('GRID') or 3D ('Lattice') using metal blocks with uniformly spaced apertures or multileaf collimators (Meigooni *et al* 2006, Ha *et al* 2006, Buckey *et al* 2010, Neuner *et al* 2012, Yan *et al* 2020). Early-phase clinical SFRT trials has shown promising results with local control rates as high as 80% and acceptable toxicities, although some of these studies have included cases also treated with conventional RT (Huhn *et al* 2006, Peñagaricano *et al* 2010, Mohiuddin *et al* 2014, Edwards *et al* 2015).

As with conventional RT, SFRT will cause tumor cell death through direct radiation-induced damage, especially in the high dose regions. Still, significant parts of the tumor will receive low doses, which from a standard radiobiological viewpoint cannot result in tumor control. However, it has recently been demonstrated that radiation also can kill cancer cells via immunomodulation (Formenti and Demaria 2013). Radiation can induce an immunostimulatory form of cell death process called immunogenic cell death (ICD), which results in activation of cytotoxic T-cells (CTLs) to explicitly target cancer cells, including non-irradiated cancer cells. SFRT could thus be used to induce immunogenic signaling in tumor cells irradiated with high doses and at the same time spare immune cells residing in the low-dose regions. Partial (or hemi-) irradiation of breast carcinoma (BC) tumors in immunocompetent mice has shown increased infiltration levels of CTLs ($CD8^+$ T cells in particular) compared to fully irradiated tumors (Markovsky *et al* 2019). Partial irradiation also caused a tumor growth delay comparable to that following whole-tumor irradiation with the same dose. Similar results have been reported for bilaterally implanted Lewis lung carcinoma (LLC) in immunocompetent mice (Kanagavelu *et al* 2014).

To establish a better quantitative evaluation of the immunological responses in conjunction with partial irradiation or SFRT, computational modelling of tumor immune response could be a valuable tool. Mathematical models of tumor regression following RT and brachytherapy have previously been developed (Lim *et al* 2008, Chvetsov *et al* 2009, Huang *et al* 2010, Arnesen *et al* 2017), taking classical radiobiological parameters such as radiosensitivity and proliferation rate into account. However, to our knowledge, no modelling studies have incorporated immunomodulation and its dependency on irradiation dose as well as spatial fractionation pattern. The purpose of the current study was to develop a computational tumor model to predict anti-tumor immunity from both conventional RT and SFRT. The tumor model is initialized by well-established parameters for breast and lung cancer cells, including radiosensitivity, proliferation, decay and clearance rate, while the immunogenic tumor response is assumed to be regulated upon irradiation. The model is subsequently fitted to two experimental datasets from independent studies using immunocompetent mouse models where the irradiation patterns have been varied. Hence, the current tumor model, encompassing the combined effects of anti-tumor immune response and irradiation, may lay a foundation for evaluating and optimizing new treatment strategies in future pre-clinical and clinical SFRT.

1.1. Immunomodulation by localized radiotherapy

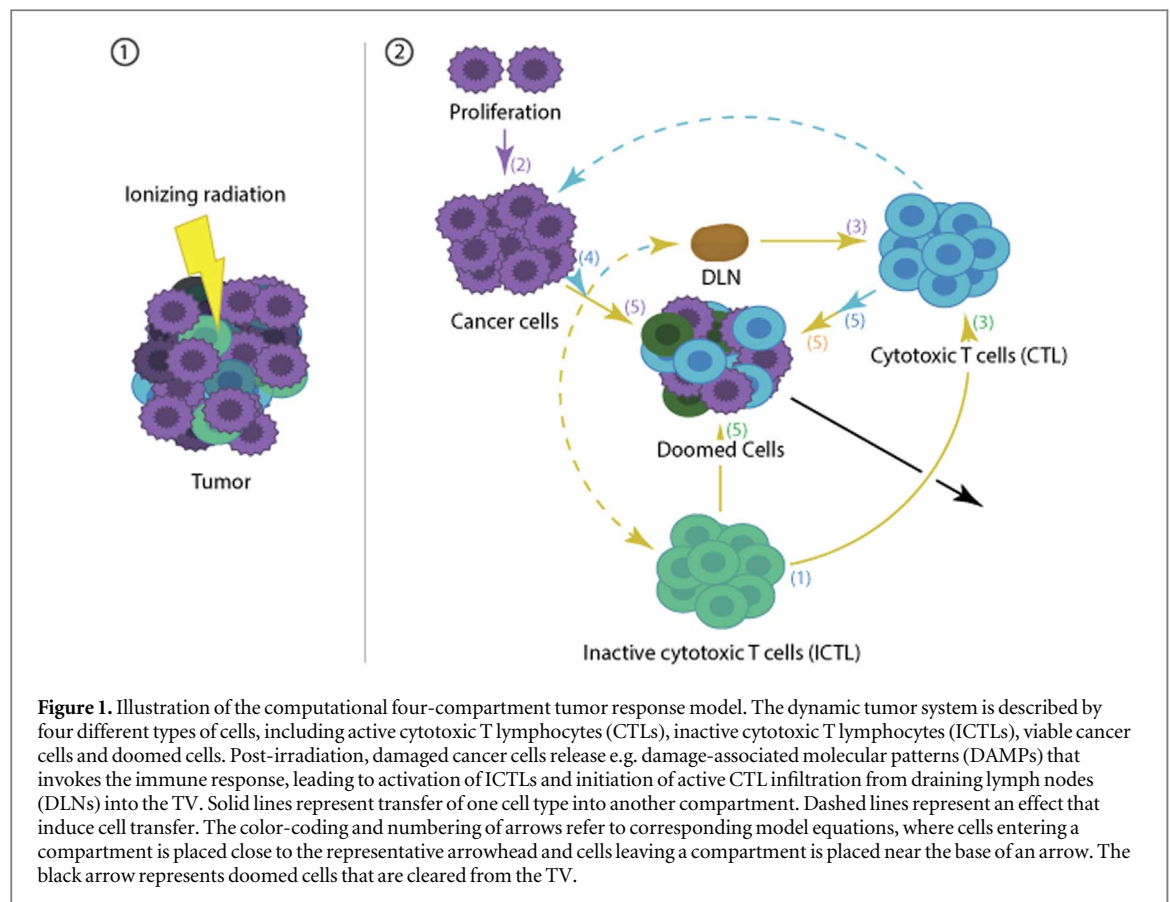
Several studies indicate that localized RT may produce systemic immune-mediated anti-tumor effects through ICD (Formenti and Demaria 2009, Garg *et al* 2015, Golden and Apetoh 2015). ICD leads to release of tumor-associated antigens and damage-associated molecular patterns (DAMPs). In addition, the production and release of cytokines and chemokines in the tumor microenvironment (TME) initiates chemotaxis of dendritic cells (DCs), macrophages and CTLs into the microenvironment through a cascade of stimulatory anti-tumor survival events (Demaria *et al* 2005, Formenti and Demaria 2013, Weichselbaum *et al* 2017). DAMP secretion leads to maturation of tumor-associated DCs into effective antigen-presenting cells (APCs). Then, the mature DCs are mobilized into draining lymph nodes (LNs) in response to chemokine secretion. In the secondary lymphoid organs (LNs and spleen), the tumor antigens are then cross-presented by the APCs to the appropriate T-cell receptor (TCR) on T cells, thus priming the TCR which subsequently leads to T-cell activation (Nolz 2015, James *et al* 2018). The antigen-specific effector T cells (e.g. cytotoxic $CD8^+$ T cells) can then exit the LNs to reach the circulatory system and institute tumor infiltration for tumor-cell-killing or other effector functions (Oelkrug and Ramage 2014). In addition to DAMP signaling, irradiation can activate the innate immune system through the presence of tumor cell DNA fragments in the cytosol after aberrant cell division (Woo *et al* 2014).

Naïve (inactive) T-cells may be present in the tumor prior to RT delivery. This reflects the tumors' ability to evade detection through recruitment of immunosuppressive cells into the TME such as regulatory T cells, tumor-associated macrophages, and myeloid-derived suppressor cells. However, RT-induced release of chemokines and tumor neoantigens could induce effective immune reactivity through enhanced priming of both pre-existing or newly infiltrated T cells. Thus, enhanced tumor cell kill may result from T-cell accumulation prior to RT with subsequent radiation-induced activation (Characiejus *et al* 2011, Liu *et al* 2012, Fridman *et al* 2012).

2. Method

2.1. Model description

The computational four-compartment tumor model (see figure 1) describes tumor growth with an anti-tumor immune response induced by RT, which involves both enhanced priming of inactive T cells already present in the tumor into CTLs and delivery of CTLs from draining LNs. Consequently, CTLs are described by two compartments in the model: active (primed) and inactive (unprimed; naïve) CTLs (ICTLs). The two remaining



compartments comprise proliferating viable cancer cells, which are affected by RT and the interplay with CTLs, and doomed cells which represents dead CTLs and cancer cells.

The developed tumor model builds on the two-compartment model first introduced by Lim *et al* (2008). They described the tumor volume (TV) consisting of undamaged and reparable (viable) and irreparably damaged (doomed) cancer cells. In the current work, to include effects of the immune system following RT, the two additional T-cell compartments are added. The role of the CTLs is to infiltrate the primary tumor and destroy cancer cells. Activation of CTLs is assumed to be proportional to the release of DAMPs and therefore related to the number of cancer cells killed by the irradiation. Immediately after irradiation, ICTLs are activated and the diffusion of CTLs from draining LNs is initiated with a continuous invasion into the tumor. CTLs killed by radiation, or inactivated due to a limited life span, are transferred to a doomed cell compartment and may reside in the tumor for some time before being cleared from the volume. Doomed cancer cells, destroyed by either radiation or CTLs, inhabit the same doomed cell compartment.

The iterative tumor model is described by five equations (equations (1)–(5)), where the number of cells in each compartment is calculated at each day i . Model symbols and parameters along with their adopted values from corresponding references are listed and explained in table 1. The different cell types are assumed to be homogeneously distributed in the tumor. A key parameter is the TV irradiation fraction (v_{TIF}), which controls the fraction of the tumor is irradiated. In the model, the surviving fraction (SF) of cells after irradiation is based on the generalized linear–quadratic (LQ) model (Fowler 1989). Together with the ICTL response to irradiation (equation (1)), viable cancer cell kill is also included in the model (equation (2)). Two equations for the viable cancer cells are needed (equations (2) and (4)), as this cell pool is reduced by both the irradiation and the action of CTLs. However, the latter comes as a consequence of immune activation, which is initiated by the transformation from ICTL to CTL and subsequent cancer cell kill. Both pre-existing CTLs and activated ICTLs due to irradiation (equation (3)) may kill remaining viable cancer cells (equation (4)). The doomed cell compartment is composed of destroyed cancer cells and CTLs due to either cytotoxic effect on viable cancer cells, general cell kill from ionizing radiation, or limited lifetime in case of CTLs (equation (5)). The TV is calculated as the sum of the volumes of the different cells residing in the tumor. The calculations for succeeding day $i + 1$ is performed in the sequential order of equations (1)–(5) and then estimated by summing equations (1), (3)–(5) representing respective compartment percentages of the TV at preceding day i . Biological specificity of the tumor immune system is ensured through fixed parameters, whereas the free parameters are optimized via model fitting (see table 1).

Table 1. List of model variables, equations, free parameters, fixed parameters and model symbols applied for fitted and predicted tumor growth simulations, with corresponding references.

Variable	Description	Equation		
V_{ICTL}	Volume percentage of inactive CTLs	Equation (1)		
V_{CTL}	Volume percentage of active CTLs	Equation (3)		
V_C	Volume percentage of viable cancer cells	Equations (2) and (4)		
V_D	Volume percentage of doomed cells	Equation (5)		
Equation symbol	Description	Equation		
SF_{CTL}, SF_C	Surviving fraction with compartment-specific α, β	$e^{-\alpha \cdot d \left(1 + \frac{d}{\alpha/\beta}\right)}$		
ϵ	Activation function for ICTLs	$\tanh(s(1 - SF_C)v_{TIF} \cdot V_C^0)$		
p	Proliferation fraction for a time interval Δt	$e^{\frac{\Delta t \ln 2}{T_{pot}}}$		
c_{CTL}	Decay fraction of CTLs for a time interval Δt	$e^{-\frac{\Delta t \ln 2}{\tau_{CTL}}}$		
c_D	Clearance fraction of doomed cells for a time interval Δt	$e^{-\frac{\Delta t \ln 2}{\tau_D}}$		
Free parameter	Description	Boundaries	Unit	References
T_{pot}	Potential doubling time for viable cancer cells without presence of CTL	$\in[1, 30]$	Days	Fitted to experimental data (Markovsky <i>et al</i> 2019)
V_C^0	Percentage of TV consisting of viable cancer cells at day 0	$\in[10, 98]$	%	Fitted to experimental data (Markovsky <i>et al</i> 2019)
V_{CTL}^0	Percentage of TV consisting of CTLs at day 0	$\in[0, 20]$ subjected to $V_{CTL}^0 + V_C^0 \leq 100$	%	Fitted to experimental data (Markovsky <i>et al</i> 2019)
k	Average viable cancer cell kill per activated CTL per day	$\in[0, 24]$	Kills/CTL/day	Fitted to experimental data (Markovsky <i>et al</i> 2019)
ω_{TIF}	Infiltration rate of CTLs per day. Subscript <i>TIF</i> indicate dependency on irradiation fraction	$\in[0, 1]$	No unit	Fitted to experimental data (Markovsky <i>et al</i> 2019)
s	Steepness parameter in the activation function ϵ	$\in[0, 100]$	No unit	Fitted to experimental data (Markovsky <i>et al</i> 2019)
Fixed parameter	Description	Value	References	
$\alpha_{CTL}, \beta_{CTL}$	Radiosensitivity of CTLs ^a	$\alpha_{CTL} = 0.182 \text{ Gy}^{-1} \beta_{CTL} = 0.143 \text{ Gy}^{-1}$	Nakamura <i>et al</i> (1990)	
$\alpha_{BC}, \alpha_{BC} / \beta_{BC}$	Radiosensitivity of BC cells ^b	$\alpha_{BC} = 0.05 \text{ Gy}^{-1} \frac{\alpha_{BC}}{\beta_{BC}} = 4.4 \text{ Gy}$	Qi <i>et al</i> (2011)	
$\alpha_{LLC}, \alpha_{LLC} / \beta_{LLC}$	Radiosensitivity of LLC cells ^c	$\alpha_{LLC} = 0.072 \text{ Gy}^{-1} \frac{\alpha_{LLC}}{\beta_{LLC}} = 15.9 \text{ Gy}$	Tai <i>et al</i> (2016)	
τ_{CTL}	Half-life of CTLs ^a	1.7 days	De Boer <i>et al</i> (2003)	
τ_D	Half-life for clearing doomed cells from TV ^d	17.5 days	Zhong and Chetty (2014)	

Table 1. (Continued.)

Model symbol	Description	Value
i	Post-RT day superscript index	0, 1, 2,...
Δt	Time step between day i and $i + 1$	1 day
$TVIF$	TV irradiation fraction	$\in [0, 1]$
I, II	Preliminary (I) and final (II) viable cancer cell compartment	

Abbreviations: SF = surviving fraction; LQ = linear–quadratic; CLT = cytotoxic T lymphocyte; ICTL = inactive cytotoxic T lymphocyte; TV = tumor volume; RT = radiotherapy; BC = breast carcinoma; LLC = Lewis lung carcinoma.

Fixed parameter specification

^a Effector CD8⁺ T lymphocyte.

^b Breast adenocarcinoma.

^c Non-small-cell lung carcinoma.

^d Head and neck cancer.

$$V_{ICTL}^{i+1} = ((1 - v_{TIF}) + v_{TIF} SF_{CTL})(1 - \epsilon) \cdot V_{ICTL}^i \quad \text{Inactive CTLs} \quad (1)$$

$$V_{C,I}^{i+1} = ((1 - v_{TIF}) + v_{TIF} SF_C) p \cdot V_{C,I}^i \quad \text{Viable cancer cells, I} \quad (2)$$

$$V_{CTL}^{i+1} = (((1 - v_{TIF}) + v_{TIF} SF_{CTL}) \cdot V_{CTL}^i + \epsilon \cdot V_{ICTL}^{i+1} + \omega_{TIF} \cdot V_{C,I}^{i+1}) c_{CTL} \quad \text{Active CTLs} \quad (3)$$

$$V_{C,II}^{i+1} = V_{C,I}^{i+1} - k \Delta t \cdot V_{CTL}^{i+1} \quad \text{Viable cancer cells, II} \quad (4)$$

$$V_D^{i+1} = \left(V_D^i + (V_{ICTL}^{i+1} - V_{ICTL}^i) + \left(\left(\frac{1}{c_{CTL}} - 1 \right) \cdot V_{CTL}^{i+1} + (1 - SF_{CTL}) v_{TIF} \cdot V_{CTL}^i \right) + (k \Delta t \cdot V_{CTL}^{i+1} + (1 - SF_C) v_{TIF} \cdot V_{C,II}^i) \right) c_D \quad \text{Doomed cells.} \quad (5)$$

A description of each equation is given below:

- (1) The first term inside the main parenthesis describe the proportion of unirradiated ICTLs that stays non-activated, while the second term represents the fraction of remaining T cells in the compartment that survive after irradiation without being activated. Hence, the CTL SF, conformed to the LQ model, is only utilized in the second term.
- (2) In the viable cancer cell compartment, the first term in the main parenthesis gives the fraction of viable cancer cell that are not irradiated and thus undergo exponential division. The second term addresses the survival of cancer cells in the irradiated part of the TV. The sum of these terms regulates the computed total number of viable cancer cells at preceding day i , which in turn is proportional to the rate of cancer cell proliferation.
- (3) The three terms within the primary parenthesis determine the number of active CTLs. The first term describes the surviving CTLs, both unirradiated and irradiated, whereas in the second term, pre-existing ICTLs in the TV are activated according to delivered dose and irradiated TV fraction. Diffusion of CTLs from draining LNs directly into the TV compartment is facilitated by the third term, where the infiltration is proportional to the number of viable cancer cells. The number of CTLs are concomitantly decreasing between preceding day i and succeeding day $i + 1$ due to their limited half-life.
- (4) After recruiting CTLs to the TV, viable cancer cells that have survived the irradiation, expressed by (2), may become inactivated by CTLs. The final number of viable cancer cells is therefore adjusted due to the targeted destruction from CTLs.
- (5) Four terms in the main parenthesis are used to represent inactivated, doomed cells from each compartment. While the first term is the pool of doomed cells from the preceding day i , the second term represents doomed ICTLs. CTLs that are doomed due to a short active half-life and radiation damage are given by the third term. Cancer cells that are killed by either CTLs or irradiation are included through the fourth term. Collectively, the number of doomed cells is being cleared from day i to $i + 1$ with an exponential rate from the TV.

2.1.1. Parameter boundaries

Free parameters are given generous boundaries in the fitting to allow for a large variety of scenarios, centered on values found in literature:

- Potential doubling time for viable tumor cells, T_{pot} , is taken from modelling of studies of tumor growth (and reference therein), where values range from 3 to 30 d (Huang *et al* 2010, Chvetsov 2013, Zhong and Chetty 2014, Arnesen *et al* 2017).
- The CTL kill rate, k , is acquired from two studies where the average kills per day was 2–24 (Vasconcelos *et al* 2015, Halle *et al* 2016). CTLs kill by direct contact with the target cell and may kill multiple targets simultaneously and in rapid succession throughout their activated lifespan (Wiedemann *et al* 2006, Halle *et al* 2017). This indicates that CTLs are highly efficient, though the frequency of each kill mechanism remains unknown.
- The infiltration rate, ω_{TIF} , is to our knowledge not available in the literature. Thus, a complete range of ω_{TIF} 's is included, where lower boundary was set to zero. The upper represents a scenario where the number of infiltrating CTLs in one day is equal to the number of cancer cells in the tumor.
- Values for V_C^0 are based on studies on viable cancer cells in untreated tumors, ranging from 13% to 92% (Panayiotou *et al* 2015, Peng *et al* 2018).

- For V_{CTL}^0 , Peranzoni *et al* (2018) found that a CTL-enriched lung tumor had roughly 361 CTLs per mm^2 . By assuming that diameter of all tumor cells is $20\text{ }\mu\text{m}$ (Jiang *et al* 2017), V_{CTL}^0 is roughly 14% (upper boundary). The lower boundary for V_{CTL}^0 is set to 0 to allow for immunocompromised tumors.
- The activation function, ϵ , of the immune system signifies activation of ICTLs, which is assumed to occur after tumor irradiation. The function is presumed to increase hyperbolically to 1 with increasing steepness parameter, s , where a high s implies substantial immune activation even at low doses. Therefore, the upper boundary (when $s \rightarrow \infty$) is considered as an extreme situation where one destroyed cancer cell may activate all ICTLs (i.e. $\epsilon \rightarrow 1$), while the lower boundary (when $s = 0$) represents a scenario where no ICTLs become activated regardless of the amount of destroyed cancer cell (i.e. $\epsilon = 0$).

2.1.2. Model assumptions

Initial conditions were established for $i = 0$ (day 0; day of RT). Although V_C^0 and V_{CTL}^0 were regarded as free parameters and found through parameter optimization (see section 2.2), a constraint was introduced on the remaining cells in the tumor. By assuming that the pool of doomed cells is negligible at treatment start ($V_D^0 = 0\%$), we have $V_{ICTL}^0 + V_{CTL}^0 + V_C^0 = 100\%$. For $i > 0$, no boundary conditions were imposed to allow for unconstrained cell compartment progression, yielding a total cell compartment sum of 100% for each successive day i .

All immune cells in the model are assumed to be CTLs, where the corresponding fixed parameters in table 1 characterizes cytotoxic CD8^+ T-cells. CD8^+ T-cells are the most relevant immune cells contributing to tumor cell death in patients undergoing RT, as regions of tumor cell death have been shown to co-exist with an abundance of CD8^+ T cells in irradiated tumors and where depletion of CD8^+ T cells lead to rapid tumor regrowth (Liang *et al* 2013). Hence, CD8^+ T cells are actively mediating the radiation induced anti-tumor immune response. Here, it is assumed that each CD8^+ T cell results in a mean cancer cell kill rate per day. The limited lifetime of these T cells is also included.

We are applying our model to murine 67NR BC and LLC tumors (see below), where intrinsic radiosensitivities are not available in the literature. Therefore, the BC tumor is simulated with a radiosensitivity acquired from an early-stage breast adenocarcinoma study that fitted the LQ-model to outcome data from a series of randomized clinical trials of RT-treated patients (Qi *et al* 2011). Employed radiosensitivity of the LLC tumor is derived from a tumor control probability modelling study based on early-stage non-small-cell lung carcinoma data (Tai *et al* 2016). The adopted radiosensitivities for both cell lines conform largely with a review study of published LQ parameter values for human tumors after clinical RT (van Leeuwen *et al* 2018) and was therefore considered representative for our tumor simulations. The clearance rate of doomed cells was taken from a previous tumor modeling study for head and neck cancer patients (Zhong and Chetty 2014).

Furthermore, tumor growth in the absence of the immune system and with no treatment was assumed to follow simple exponential growth that is proportional to the number of cells present and is captured by a single growth rate constant.

2.2. Free parameter optimization

The tumor response model is tuned by optimizing all free parameters (see table 1) to experimental syngeneic xenograft data (Markovsky *et al* 2019). In that study, partial and full tumor irradiation was investigated in immunocompetent BALB/c mice bearing orthotopic 67NR BC tumors. The mean TV was 250 mm^3 at the time of treatment. For the irradiation experiment, 3 treatment groups with up to 54 mice per group was used: (1) no RT (controls), (2) 50% irradiation, and (3) 100% irradiation. For partial (50%) or full (100%) irradiation, the mouse was positioned so that either half or whole tumor was located inside the radiation field, respectively. The mice received a single 10 Gy dose at day 0 by either partial or full irradiation. The effect of an anti- CD8^+ antibody was also studied in 5 animals. A total of $n = 114$ observations were lifted from the publication.

We also performed model optimization for the LLC tumors in C57BL/6 mice used in the same study (Markovsky *et al* 2019). The LLC line is considered less immunogenic compared to 67NR. The tumor model was fitted to experimental LLC data obtained from the same irradiation conditions as the 67NR tumors. However, the mice received a single dose of 15 Gy at day 0 by either partial or full irradiation to the tumor. A total of $n = 43$ observations were lifted from the publication.

Employing the sampled experimental data, the model for each cell line was optimized separately allowing for cell line-specific estimates of the free parameters. First, T_{pot} was independently fitted to the anti- CD8^+ (without RT) data in order to describe the potential cancer cell doubling time for CD8^+ T-cell depleted tumors. It was here assumed that the whole tumor only consisted of viable cancer cells ($V_C^0 = 100\%$). Then, adopting the estimated T_{pot} , the remaining free model parameters were fitted simultaneously to control, 50% and 100% irradiation data. In the model fitting, a least squares method for parameter estimation was developed (see supplementary document (available online at stacks.iop.org/PMB/66/175012/mmedia)).

All sampled TVs were normalized to unity at day 0. The optimization performance was evaluated using the root-mean-squared error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \tilde{y}_j)^2},$$

where n is the number of sampled experimental points, y_j is the j th sampled experimental measurement and $\tilde{y}_j$ is the corresponding fitted value.

Uncertainty in the model optimization was introduced by lifting the standard error of the mean σ_m from the observations in the given publication (Markovsky *et al* 2019). In that paper, σ_m was approximately 150 cm³, irrespective of the absolute TV. In the current work, synthetic TV datasets were generated by random sampling from Gaussian distributions having means equal to the reported values and a standard deviation of 150 cm³. In total, 5000 synthetic datasets were generated for both 67NR and the LLC tumors. Each synthetic dataset was then subjected to the least squares optimization, providing 5000 model fits and thus incorporates variability in the TV data. From this, 95% confidence intervals (CIs) in the estimated model and corresponding parameters were obtained. Furthermore, a sensitivity study of the model parameters was conducted for the 67NR dataset in order to assess the impact on the tumor response from each parameter (see supplementary document).

All analysis and modelling were performed by in-house software developed in Python 3.6.

2.3. Model prediction

Predictive capabilities of the tumor response model was tested on experimental data for 67NR tumors receiving higher doses than in the optimization dataset (Markovsky *et al* 2019). The proposed tumor model was attempted validated on data from two additional experiments using 15 and 20 Gy single doses to mice bearing 67NR tumors were employed. The sampled data was based on mean TV data measurements for irradiation with 15 and 20 Gy of 50% ($n = 17$ and $n = 19$ observations, respectively) or 100% ($n = 17$ and $n = 10$ observations, respectively) of the TV.

Additionally, model predictions were performed for animals with 67NR tumors receiving 10 Gy as a single dose in combination with FTY720; an immunomodulating drug that inhibits lymphocyte production in lymphoid tissues (Brinkmann *et al* 2010). The various cancer-associated mechanisms of FTY720 were taken into consideration in the simulated predictions.

The model predictions for the FTY720 experiments were included in the Discussion, as these findings were less forward to interpret. Tumor growth predictions were also conducted on a second LLC tumor growth dataset (Kanagavelu *et al* 2014) (see supplementary document).

The prediction performance was assessed using the mean absolute percentage error (MAPE):

$$MAPE = \frac{1}{n} \sum_{j=1}^n \left| \frac{y_j - \tilde{y}_j}{y_j} \right|,$$

where n is the number of sampled experimental points, y_j is the j th sampled experimental measurement and $\tilde{y}_j$ is the corresponding predicted value.

3. Results

3.1. Free parameter optimization of 67NR BC data

The tumor model was fitted to the sampled experimental data of 67NR tumor growth delay (Markovsky *et al* 2019) by optimizing the free parameters summarized in table 1. First, the unirradiated group with anti-CD8⁺ injection was used to independently acquire an estimate of the viable cancer cell potential doubling time, T_{pot} . Then, the model was simultaneously fitted to the datasets for the three remaining irradiation experiments. Qualitatively, the model, with 95% CIs estimated from the random sampling procedure, fitted the data quite well (figure 2). Corresponding parameters and their uncertainties together with RMSEs are listed in table 2. The lowest RMSE was observed for the anti-CD8⁺ group, while the unirradiated (control) group resulted in the highest deviation owing to a larger scatter of data points between 2 and 9 d post-RT. For both irradiation patterns (50% and 100% irradiation), the model fit showed good agreement with experimental data, though fewer sampled observations were available at day 15 post-RT.

The underlying temporal dynamics of each tumor compartment for 50% and 100% irradiation are shown in figure 3. A discontinuity is evident in all compartments at day 1 post-RT representing the response to the single 10 Gy dose delivery. A larger portion of viable cancer cells are seen to be killed due to 100% irradiation at day 1. With no pre-existing effector T lymphocytes within the TV at day 0 ($V_{CTL}^0 \approx 0\%$), CTL accumulation is evident the first 10 d after treatment start, where partial irradiation demonstrated a quicker and higher build-up

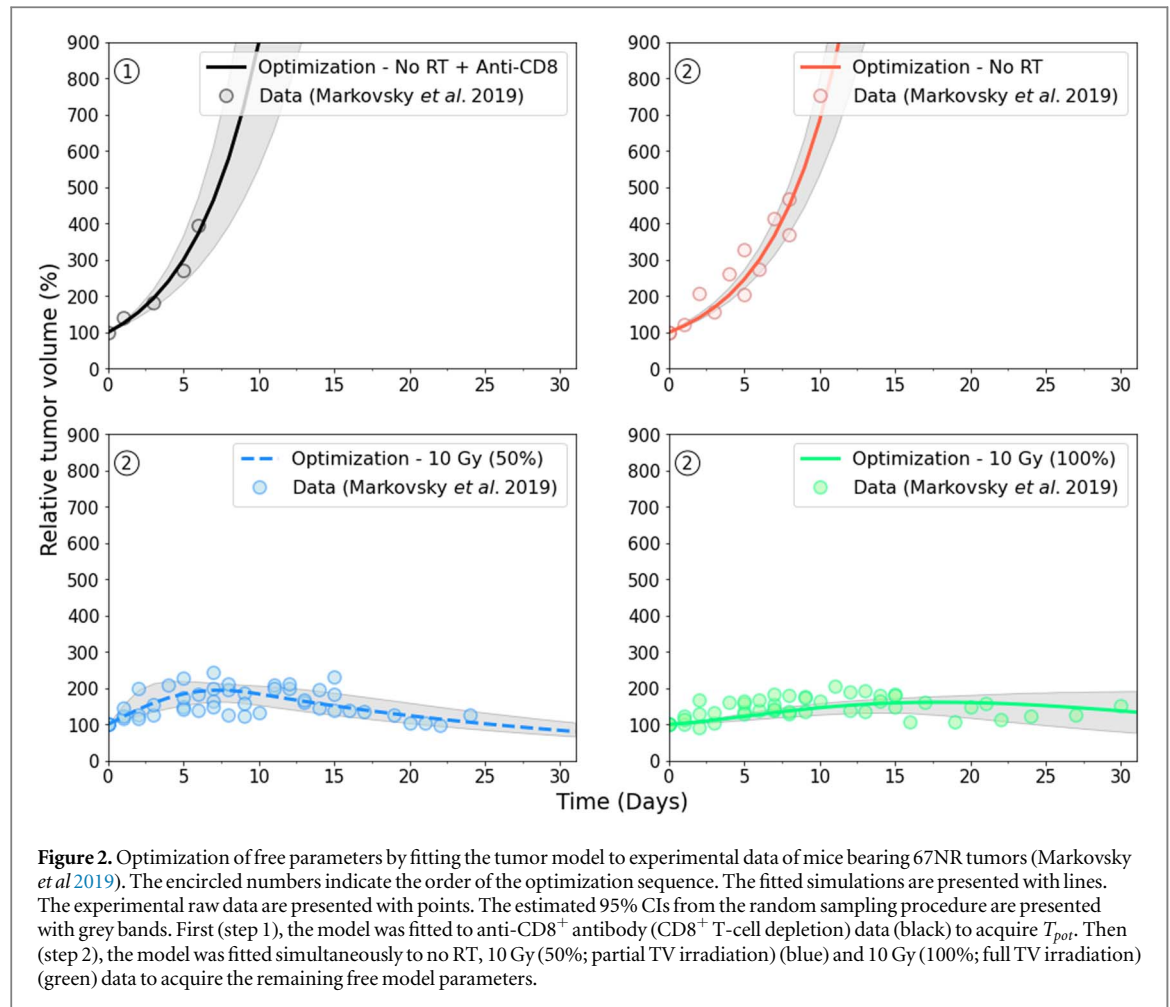
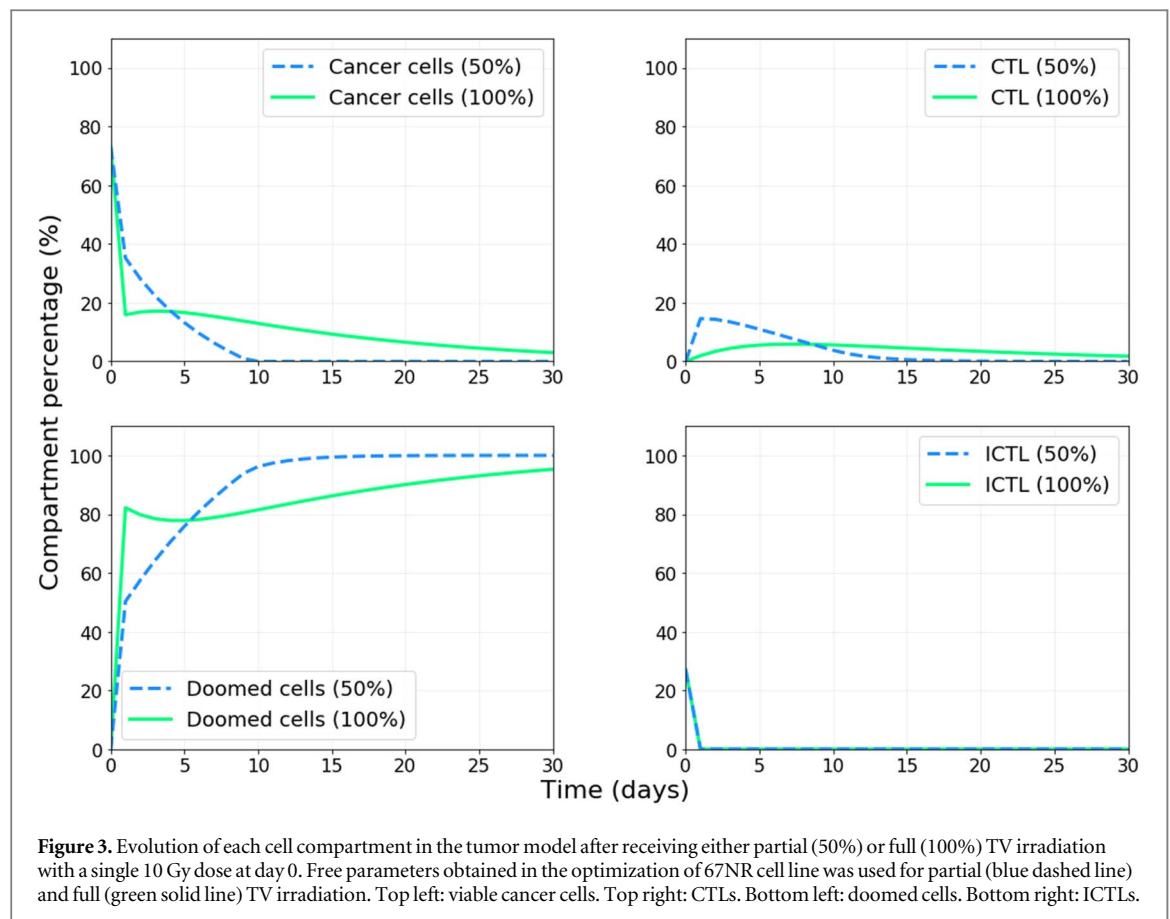


Table 2. Optimized parameters by fitting the model to experimental 67NR and LLC cell line datasets. Parameter estimates are presented together with 95% CIs estimated from the random sampling procedure. The subscript 50% and 100% indicates partial and full TV irradiation, respectively. The performance of the optimization is evaluated with RMSE for each simulated cell line.

Fit parameter	67NR	LLC
	Estimate (95% CI)	Estimate (95% CI)
T_{pot} (Days)	3.20 (2.68, 4.06)	3.70 (3.45, 4.12)
V_C^0 (%)	73.0 (55.4, 97.9)	97.1 (83.9, 97.9)
V_{CTL}^0 (%)	$1.3 \cdot 10^{-6}$ ($4.0 \cdot 10^{-7}$, 19)	$9.1 \cdot 10^{-6}$ ($2.4 \cdot 10^{-7}$, 0.62)
k (Kills/CTL/day)	0.66 (0.39, 8.30)	24.0 (3.0, 24.0)
$\omega_{50\%}$ (No unit)	0.24 (0.017, 0.81)	$1.5 \cdot 10^{-6}$ ($1.0 \cdot 10^{-8}$, 0.0089)
$\omega_{100\%}$ (No unit)	0.17 (0.014, 0.33)	$9.6 \cdot 10^{-4}$ ($4.0 \cdot 10^{-8}$, 0.99)
s (No unit)	0.7 (0.0001, 14.2)	0.9 (0.04, 56.7)
RMSE		
RMSE—No RT + anti-CD8 ⁺	17.7	14.3
RMSE—No RT	46.7	44.4
RMSE—50% irradiation	31.2	22.4
RMSE—100% irradiation	28.0	18.1
Total RMSE	65.2	54.8

reaching a maximum of approximately 18% TV occupancy around day 2 post-RT compared to 9% for whole tumor irradiation near 8 d post-RT. This is also explained through a higher estimated CTL infiltration rate for 50% partial TV irradiation, $\omega_{50\%}$, listed in table 2. The difference between partial and total irradiation is mainly due to the high radiosensitivity of CTLs as well as a higher estimated infiltrate of CTLs from draining LNs. Proportionately with the increasing trend of CTLs, a more rapid reduction in cancer cells (from $V_C^0 = 73.0\%$) and a larger pool of doomed cells was observed on day 1 post-RT for partially irradiated tumors. All ICTLs

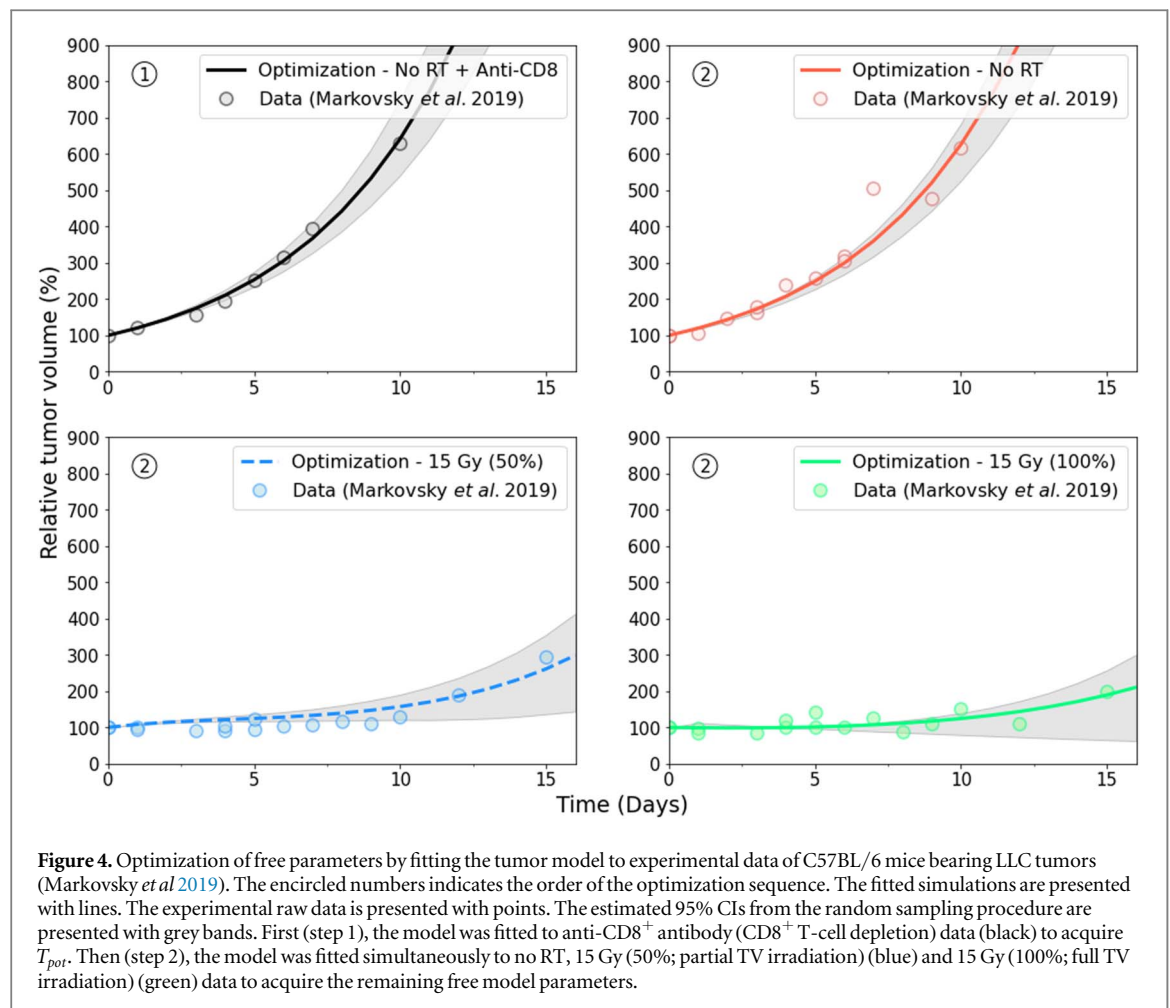


($V_{ICTL}^0 = 27.0\%$) became activated during the first day of treatment regardless of irradiation pattern. After depletion of ICTLs, kinetics of active CTLs was primarily governed by the exponential decay with a rate of $\ln 2/\tau_{CTL}$ (see table 1). Eventually, the entire TV composition is converging towards containing exclusively non-proliferating doomed cells for both irradiation patterns.

3.2. Free parameter optimization of LLC data

The tumor model was also fitted to experimental data of LLC tumor growth data (Markovsky *et al* 2019). All model fits resulted in small deviations from the experimental mean values with rather narrow 95% CIs (figure 4), where free model parameters with 95% CIs and RMSEs are listed in table 2. The lowest and highest RMSE was calculated for the anti-CD8⁺ and control experiment, respectively. Although a prolonged growth delay was observed for both partial and full TV irradiation when compared to the unirradiated control group, persistent tumor growth towards day 15 post-RT was proposed by the fitted model curves, where a slightly greater tumor progression was observed for partially irradiated tumors. However, the estimated 95% CIs indicate that the predicted tumor progression after about day 10 is highly uncertain.

The temporal dynamics of each cell compartment for LLC tumors subjected to partial and full TV irradiation is shown in figure 5. The impact of the single dose of 15 Gy on each tumor compartment is demonstrated at day 1 post-RT, where a greater fall-off in viable cancer cells is seen for full TV irradiation. This causes an equivalent increase in the doomed cell compartment. For the CTL population, a slightly higher accumulation was seen for partially irradiated tumors relative to fully irradiated tumors within the first 6 d post-RT, reaching a maximum of 1% TV occupancy. Quantitatively, since $V_{CTL}^0 \approx 0\%$ and $V_C^0 = 97.1\%$ (see table 2), the initial portion of ICTLs in the TV was calculated to 2.9%. This portion may potentially become activated upon irradiation. According to the model fit, little CTL infiltration was induced for either irradiation method ($\omega_{50\%}, \omega_{100\%} < 10^{-3}$). Together with the fact that the CTLs are simulated with a high cancer kill rate ($k = 24.0$ cancer cell kill per CTL per day), the inhibition of tumor growth until day 10 post-RT was mainly due to direct radiation effects and a radiation-induced immune response of kill-efficient CTLs that were already present in the tumor before treatment. Thus, low infiltration rate and insufficient priming of CTLs resulted in high tumor cell survivability and subsequent tumor growth for both partially and fully irradiated tumors.



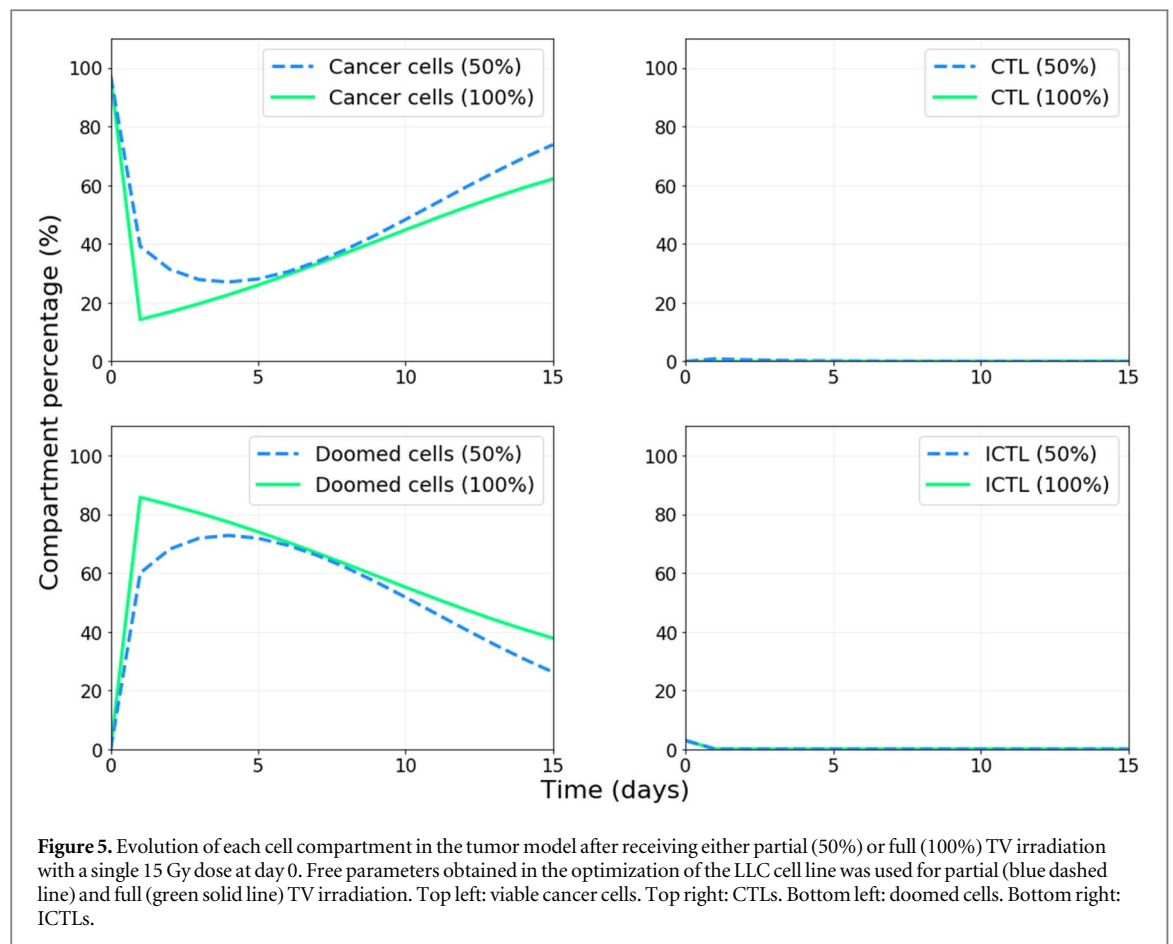
3.3. Model prediction

Using the previously optimized parameters, predictive capabilities of the tumor model was tested on different experimental datasets for 67NR tumors (Markovsky *et al* 2019). The predictions were made for four different irradiation experiments (see figure 6) together with 95% CI bands, where a single dose of 15 or 20 Gy was given by partial or whole tumor irradiation. For both irradiation methods, and irrespective of dose, our tumor response model predicted tumor regression, whereas experimental data indicated slight regrowth. This divergence is particularly observed for 15 Gy dose. As a consequence, the mean relative deviations were 24% and 34% for 15 Gy, partial and full irradiation respectively, and 26% and 19%, respectively, for 20 Gy.

4. Discussion

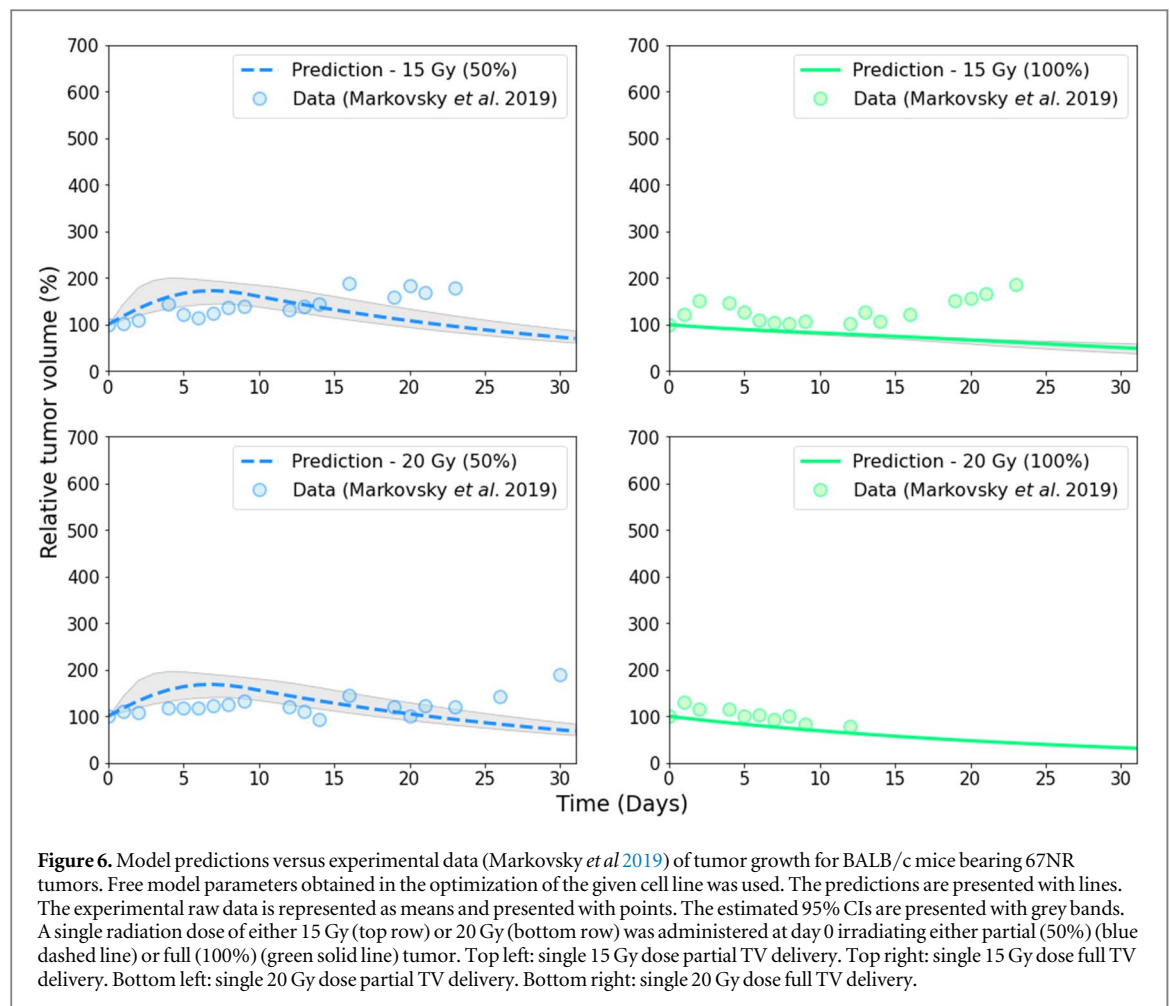
4.1. Model fit

By mathematical modelling of the cancer immune response and its synergy with RT, a tumor growth model consisting of four different compartments (inactive CTLs, active CTLs, proliferating cancer cells and doomed non-proliferating cells) was developed and rendered. We have built a tumor model which incorporates 12 parameters in total, with 6 free parameters (viable cancer cell potential doubling time T_{pot} , volume percentage of viable tumor cells at day 0 V_C^0 , volume percentage of CTLs at day 0 V_{CTL}^0 , mean viable cancer cell kill per day k , CTL infiltration rate ω and activation sensitivity s). The remaining parameters were obtained from relevant literature. The simulated temporal kinematics and interactions between the tumor cell compartment gave a satisfactory prediction of tumor response to various forms of RT delivery. We also included a random sampling procedure to estimate uncertainties in the model and free parameters. The model CIs were rather narrow, but some parameter CIs were substantial. It should be noted that the proposed model, including terms involving summation of several exponential factors, may potentially give unrealistic estimates (Chvetsov *et al* 2015). Still, our parameter estimates were mostly within ranges reported in the literature (confer section 2.1.1 and table 2), indicating that the model is rather robust. Moreover, the risk of losing predictive power due to excessive number of free model parameters was reduced through a low ratio of free model parameters to total number of



parameters (6/12), without oversimplifying the radiobiological nature of the tumor and immune response. This gives confidence in the model's capability to predict outcomes in different independent experimental scenarios whilst avoiding modelling results resembling experimental data because of overfitting. The developed tumor model may be extended to include additional parameter components to provide a greater complexity and a theoretically better description of the immune system, for instance DCs, multiple T-cell sub compartments, immunosuppressive cells, the interaction rate between tumor cells and CTLs or expression of inhibitory/stimulatory receptors/proteins responsible for CTL-function. However, maintaining a low free-to-total model parameter ratio aids assessment of the main factors determining the response of the tumor system. Several of the free model parameters may be determined in an experimental setting. For instance, cell kinetic analysis by flow cytometry can aid estimation of T_{pot} (Begg *et al* 1985). The parameters V_C^0 and V_{CTL}^0 may be estimated by histopathological and immunohistochemical assessment. ω can be determined by implanting tumor cells into a mouse model and then determine the CTL density in tumor versus tumor surface area at various time points (Valentinuzzi *et al* 2019). Incorporating one or more of these parameters as fixed in the model would likely lead to a reduction in the uncertainty of the remaining free parameters.

From the sampled experimental data from literature for murine 67NR and LLC cells (Markovsky *et al* 2019), the model parameters T_{pot} , V_C^0 , V_{CTL}^0 , k , ω and s were fitted simultaneously to each of the three experimental arms (unirradiated controls, 50% TV irradiation, 100% TV irradiation) for each individual cell line. Prior to fitting the free parameters k , ω , V_C^0 and V_{CTL}^0 , a separate fit was performed on an unirradiated group receiving anti-CD8⁺ T cell treatment to find the proliferation rate T_{pot} for each cell line. It is evident that the longer T_{pot} for LLC suggests slower proliferation compared to 67NR cells ($T_{pot}^{BC} = 3.16$ d, $T_{pot}^{LLC} = 3.73$ d), although the estimated 95% CIs from the random sampling procedure are overlapping. On the other hand, the growth of LLC tumors was less affected by the immune system (evident from growth curves without anti-CD8⁺ T cell administration). Also, LLC tumors were less responsive to irradiation in general. Comparing the fitting results for the two cell lines, the total RMSE value was lower for LLC which was due to a larger spread in 67NR data, fewer experimental LLC observations and the lack of experimental data from day 15 post-RT for LLC. Furthermore, the growth data for LLC tumors indicated a higher proportion of cancer cells with few active CTLs. For 67NR tumors, the model estimated a small proportion of pre-existing intra-tumoral CTLs (both active and inactive) prior to RT as well as CTL accumulation post-RT (see figure 3). This is in line with findings in Markovsky *et al* (2019), where immunofluorescence staining of CD8⁺ T cells showed a 3-fold amplification of

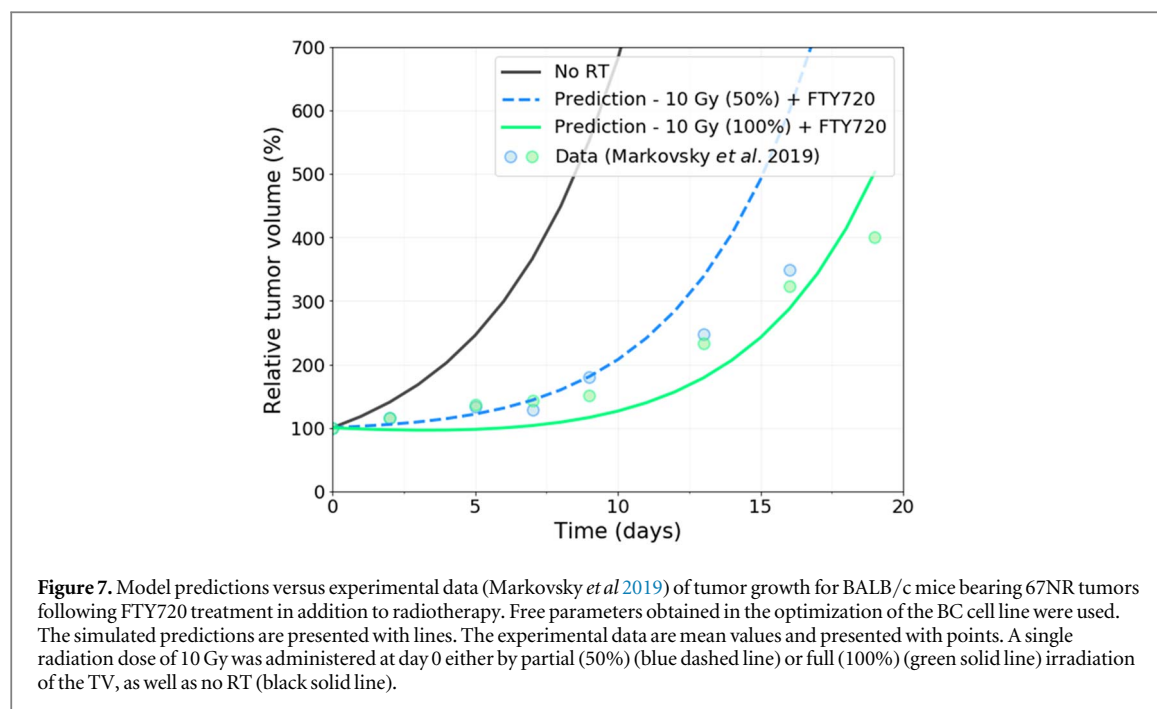


CD8⁺ cells in partially irradiated tumors at 24 h after irradiation with 10 Gy compared to full irradiation with the same dose.

To ascertain the model's competency to capture the true underlying T-cell modulation and potency upon conventional RT and GRID therapy, experimental data should be sampled from different types of tumors exhibiting high variability in immunogenicity. Since host immune response, rather than the intrinsic radiosensitivity of cancer cells, has been suggested as the key factor in determining the efficacy of RT (Liang *et al* 2013), the association between the model parameters k and ω and host immune characteristics can be explored in future studies. The tumor model indicated absence of CTLs at day 0, which may not reflect the true immunological state prior to RT. Studies have suggested that pre-treatment lymphocytic tumor infiltration, giving a spontaneous T cell response, is a prerequisite for curation (Demaria *et al* 2016, Crittenden *et al* 2018). Therefore, the V_{CTL}^0 model parameter, reflecting functionally active resident T cells at start of RT, may well be non-zero for other tumors. Moreover, with regards to CTL activation upon radiation treatment, where potentially all inactive immune cells could be recruited, an assessment of the steepness parameter, s , should be done in order to investigate the responsiveness of the T-cell activation function, ϵ , with respect to dose and irradiated TV. Especially, the activation sensitivity, represented by s , should be inspected for tumors subjected to RT in the low-dose region with partial and whole volume irradiation since our model optimizations were conducted on experiments subjected to doses ≥ 10 Gy. Also, ϵ could be modified with a dependency on remaining viable cancer cells after irradiation to reflect a signaling response to effector T cells.

A recent study found that full irradiation of murine colon tumors in BALB/c mice with single 30 Gy dose rapidly ignited the infiltration of CD8⁺ tumor-killing T cells and thereby altered the immunosuppressive microenvironment in the tumors. A single 15 Gy dose exhibited the same behavior but dampened (Filatenkov *et al* 2015). Interestingly, partial irradiation requires lower single doses of 10–20 Gy to reproducibly activate an anti-tumor immune response that augments the tumor control, where 20 Gy approaches a saturated immune response limit (Kanagavelu *et al* 2014, Markovsky *et al* 2019).

A precarious feature of our model fitting is the potential ill-posedness in the least square parameter estimation when the proposed tumor response model represents a complex summation of several exponential factors (Chvetsov *et al* 2015). These exponentials processes include treatment induced T-cell activation, cell kill,



tumor proliferation rate and the rate of cell decay, and can therefore lead to large errors in the fitting, ultimately producing unstable solutions. It has been shown that variational regularization can significantly improve estimates of parameters such as the cell SF and potential doubling time (Chvetsov *et al* 2015).

4.2. Model prediction

Model predictions, performed on independent 67NR experimental data (Markovsky *et al* 2019) with different irradiation schemes, were based on the previously established fitted model parameters. The predictions made were largely within experimental variation (see figure 6). Overall, the developed computational tumor model was able to predict tumor growth curves, but experimental data (15–20 Gy, partial/full TV irradiation) indicated some regrowth towards the last simulated days while the model predicted tumor cure. An explanation as to why the simulated predictions showed minor discrepancies from experimental data might stem from the fixed parameters that were employed from numerous published studies, instead of having tailor-fitted estimates for the adopted cell lines and experiments. Still, 10 Gy gave virtually complete tumor remission in the dataset used for optimization, so this could also point to variations between experiments and passages of the tumor cell lines.

Prediction of 67NR tumors with FTY720 injection was made as shown in figure 7. Although the primary action of FTY720 is inhibiting lymphocyte production, there is evidence that FTY720 is by itself a cytotoxic agent with cancer-killing capability that additionally could suppress cancer growth (Patmanathan *et al* 2015). However, Markovsky *et al* (2019) observed no treatment effect on 67NR tumors with FTY720 up to 7 d. In the current work, anti-infiltration restrictions on CTLs by FTY720 was manually taken into account by setting $\omega = 0$. The cytotoxicity of FTY720 was included by multiplying all cell compartments (except doomed cells) by a factor lower than 1. Still, the predictions significantly diverged from the respective growth data, irrespective of treatment pattern (see figure 7). Initially, the prediction made for partial tumor irradiation gave an appropriate fit during the first week, but tended eventually towards overestimating the tumor growth volume. The data could point towards latent pre-existing T cells in the tumor that could become potentiated sometime after irradiation. Furthermore, with FTY720 treatment, it is expected that full tumor irradiation induces a stronger tumor response compared to partial irradiation, while the experimental data are surprisingly similar. A possible influence from FTY720 could be perturbation of signaling pathways that regulates cell cycle progression and apoptosis, leading to an increased radiosensitivity of the breast cancer cells (Marvaso *et al* 2014). Another issue not accounted for is the FTY720 elimination half-life, which is reported to be 4–9 d (Horga and Montalban 2008).

A critical aspect of our proposed model that might explain the discrepancies between model fit or prediction and data is that the computational model is governed by parameters which are assumed to be constant throughout the simulated timespan. In reality, dynamic changes due to effects induced by RT on the TME, particularly effects on the vasculature, stroma and immune system itself (Barker *et al* 2015) might modify the kinematics of the CTLs. A parameter affecting the time interval for the onset of T-cell activation is the epitope density of DCs. Such latency in stimulation could therefore lead to delay, or potentially lack, of tumor growth

inhibition. Furthermore, trafficking and infiltration of T cells into tumors is also dependent on various factors as well as checkpoint pathways (Slaney *et al* 2014). Hence, a sustained T cell infiltration from LNs into the tumor bed, as included in our model, may not be true, and a continuous regulation of ω may thus be considered. Presumably, migration of effector CTLs could be normalized through a proportional dependency of growing tumor population reflecting evolving angiogenesis (Peske *et al* 2015, Lanitis *et al* 2017). Immunosuppressive cells can also be recruited into the TME upon RT contributing to inhibition of immune infiltration and activation through various mechanisms. E.g. a rapid increase in both T_{reg}-cell and MDSC numbers (within around 3 d) has been reported in response to localized RT for various dose schemes (Kachikwu *et al* 2011, Crittenden *et al* 2012, Xu *et al* 2013), while a drastic decrease in MDSC number was observed 7–14 d after single dose deliveries (Deng *et al* 2014, Filatenkov *et al* 2015). However, due to the lacking experimental data on the quantitative impact of these aspects, they are regarded and treated as second-order errors, which are omitted from our computational model.

5. Conclusion

An iterative tumor radiation response model has been developed that incorporates essential immune system components. The model was found to be a valuable tool for assessing tumor growth and regression in response to a single dose of partial or full tumor irradiation. TV estimates were simulated with good accuracy with respect to predicting experimental data derived from a recent exploratory study on murine tumors. Further model refinement and development should be conducted, in particular if new experimental data becomes available.

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Conflict of interest

The authors have no conflict of interest to report.

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